

DIPESH TRIPATHI

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Education

University of South Dakota

Master of Science in Computer Science (AI Specialization)

Aug. 2024 – Present

Vermillion, SD, USA

Tribhuvan University (Pulchowk Campus)

Bachelor in Electronic, Communication and Information Engineering

Nov 2018 - Feb 2023

Lalitpur, Nepal

Experience

Techkraft Inc

Data Engineer (US Healthcare)

Jan 2024 – Aug 2024

Lalitpur, Nepal

- Understood the workflow of US healthcare systems, including stakeholders, policies, and healthcare plans.
- Utilized SQL to query and analyze large datasets, extracting actionable insights to inform business decisions.
- Worked with AWS Cloud services, including S3, Lambda, and EC2, to manage and process large-scale healthcare data.
- Implemented data warehousing solutions using Snowflake, optimizing data storage and retrieval processes.
- Built data flow pipelines using AWS Lambda, integrating multiple data sources for healthcare data and transforming it as per business requirements.
- Developed UAT plans and created detailed distribution reports in Excel, ensuring data accuracy and integrity.
- Integrated Apache Kafka for real-time data streaming and processing, enhancing the efficiency of data ingestion and analysis pipelines.
- Automated ETL processes using Databricks Pipelines, reducing manual intervention and increasing pipeline reliability.

Teksewa

Software Engineer

Mar 2023 – Nov 2023

Kathmandu, Nepal

- Explored the core concepts of Web Development using .NET Core, Angular, and MsSQL, focusing on scalable and maintainable code practices.
- Rewrote the legacy WebForms UI using Angular, enhancing user experience with responsive design and improved functionality.
- Developed and optimized complex SQL queries and stored procedures in MsSQL to handle large datasets, ensuring efficient data retrieval and manipulation.

Projects

Proteomics Chatbot Project | *Python, FAISS, LLM (Llama3.2:1b), Docker, Streamlit, Langchain*

November 2024

- Developed a chatbot that provides detailed information about proteins by querying the UniProt database using their primary Accession ID, names, or descriptions.
- Built an interactive Streamlit-based user interface for users to ask queries and receive detailed protein data in a conversational format.
- Experimented with multiple approaches for efficient retrieval, including Sentence Transformers, ProtBERT, and FAISS indexing.
- Implemented a hybrid system combining vector search and UniProt API queries to enhance accuracy and flexibility in responding to user queries.
- Containerized the entire application using Docker, ensuring seamless deployment, scalability, and reproducibility across environments.
- **GitHub Repository:** https://github.com/Dipeshtripathi13/proteomics_chatbot

Dental Caries Segmentation Project | *Python, PyTorch, U-Net, ResNet34, Albumentations* Aug 2024 - Nov 2024

- Developed a U-Net-based model with ResNet34 encoder for segmentation of dental caries in panoramic X-ray images.
- Created and utilized a high-resolution dataset of 233 annotated X-rays, with multi-class segmentation masks for teeth, caries, and background.
- Implemented data augmentation techniques and evaluated the model using IoU and Dice metrics to achieve robust segmentation performance.
- **GitHub Repository:** https://github.com/Dipeshtripathi13/Dental_carries_Segmentation.git

Semantic Search For Nepali Document | *Python, Transformer, NLP, Angular* January 2024

- Developed a platform to upload the large document and find only relevant information by querying.
- Used Bert embedding to create the embedding vector for the document with the window size of 3 sentences and to create embedding vector for the query and used cosine similarity to find the best document window relevant to search query.
- Implemented back-end in flask and front-end with angular

Automatic Text Summarization using LSA with Transformer | *Python, Transformer, NLP* February 2023

- Implement Latent Semantic Analysis(LSA) for extractive summary and Transformer Architecture for abstractive summary of the Provided document.
- Utilized TF-IDF to generate heading of the document.
- Model has achieved the average Rouge-l score of the abstractive summary with the reference summary to be 0.422.

Sentimental Analysis of Social Media comments | *Python, Flask, React, Machine Learning* January 2022

- Designed a web framework to analyze social media comments and rate the post in term of sentiment.
- Combine Support Vector Machine(SVM) and Multinomial Naive Bayes in weighted configuration to train the modal to achieve the accuracy of 93.53%.

Technical Skills

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|---|---------------------------|
| • Programming languages(Python, C++, C#) | • NLP and Computer Vision |
| • Pyspark | • .Net |
| • SQL | • Angular |
| • Excel | • Streamlit |
| • PowerBI | • LangChain |
| • Cloud Services(AWS) EC2, S3, AWS Lambda | • Databricks |
| • Snowflake, Kafka, Airflow | • US Healthcare |

Certifications

- AWS Academy Graduate - AWS Academy Cloud Foundations (Amazon Web Services)
- AI in Healthcare Specialization (Coursera)
- ETL and Data Pipelines with Shell, Airflow, and Kafka (IBM)